From Perception to Punchline:
Empowering VLM with the Art of In-the-wild Memes

HUMOR

NeurIPS 2026

Hierarchical Understanding and Meme Optimization

Xueyan Li [*1] **Yingyi Xue** [*1,2] **Mengjie Jiang** [3] **Qingzi Zhu** [1] **Yazhe Niu** [†4]

[1] Shanghai AI Laboratory [2] Fudan University [3] Columbia Engineering, Columbia University [4] CUHK MMLab

01 What makes a caption a meme?

Same visual relation, different context-specific punchlines.

Template **Meme**



USER INPUT
Technology · "Apple fanatic"

A · Subject	B · New focus	C · Neglected
APPLE FANS	IPHONE 11	IPHONE 10

One template, many possible jokes.

How do we explore different ideas, then learn which captions work better?

Explore meanings → compare expressions

02 HUMOR: explore before seeing the answer → compare locally → update with quality control

A Two-stage reasoning supervision

1 EXPLORE · REFERENCE CAPTION HIDDEN

Infer the template's intent; explore plausible situations before seeing the answer.



- Confusing work problem → bewilderment
- Bad joke → an exaggerated painful response
- Social ignorance → refusing to listen

2 REVERSE-ANCHOR · REFERENCE REVEALED



Infer the actual context from the caption; explain why an existing path fits it.

Keep the other explored associations.

Reveal the answer only after exploration to reduce answer-driven rationalization.

Generation: template + user intent; no reference caption.

B Local preference learning

Humor is subjective; an absolute score is hard to calibrate. With the template and intended theme fixed, relative preference is easier to interpret.

Where do the training comparisons come from?

Five difficulty tiers: wrong text, wrong location, boring, detailed boring, and generated memes.

Original ≻ **Boring**





Train pair preferences with binary cross-entropy.

$$p_{ij}^{G} = \sigma[s_\phi(m_i) - s_\phi(m_j)]$$

$$\mathrm{EBC}_G(i) = \sum_{j \neq i} p_{ij}^{G}$$

Expected wins → group scores → ranking → target q

C Diversity with quality control

Exploration broadens ideas but can include weaker jokes. Preferences guide quality; KL limits drift.

A Bound the gap in normalized path humor

$$1 - \mathbb{E}_{r \sim P_\theta}[\tilde{h}_G(r)] \leq (1-\alpha)\delta$$

Best-path mass α; other paths' mean gap ≤ δ.

B Turn expected wins into local targets q

Favor captions that beat more alternatives.

C Update the policy near the CoT reference

$$\mathcal{L} = -\mathbb{E}[\sum_k q_G(m_k) \log \pi_\theta(c_k \mid I)] + \beta \Delta_{\mathrm{KL}}$$

$$H_\theta \geq H_{\mathrm{CoT}} - \sqrt{\Delta_{\mathrm{KL}}/2}$$

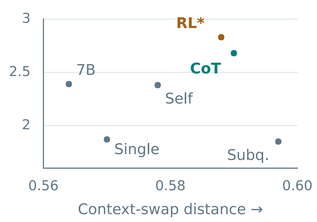
H: normalized caption humor. ΔKL: drift from CoT. Bounds on loss are conditional; diversity gains are empirical.

03 Generation quality and diversity

Model / variant	Hum. ↑	Read. ↑	Rel. ↑	Orig. ↑	Sim. ↑	Dist. ↑	Human % ↑
Qwen2.5-7B	2.39	3.35	2.91	2.57	0.576	0.564	75.7
Qwen2.5-32B	2.54	3.52	3.09	2.76	0.564	0.566	82.2
InternVL3-8B	2.39	2.79	3.04	2.79	0.545	0.564	62.7
GLM-4.1V-9B-Thinking	1.73	2.62	2.75	2.71	0.602	0.572	45.1
Keye-VL-8B-preview	2.35	3.19	2.99	2.71	0.585	0.580	69.0
GPT-4o	2.70	2.99	3.21	**2.97**	0.603	0.552	91.3
Gemini-2.5-flash	2.81	3.29	3.25	2.88	0.600	0.561	—
HUMOR-CoT	2.68	**3.70**	3.50	2.90	**0.640**	0.590	91.5
CoT: Single Path	1.87	2.79	2.68	2.45	0.637	0.570	86.0
CoT: Self-Improve	2.38	3.68	3.00	2.65	0.629	0.578	89.1
CoT: Subquestion	1.85	3.32	2.58	2.47	0.639	**0.597**	87.2
HUMOR-RL (preview)	**2.83**	3.67	**3.55**	2.79	0.631	0.588	**92.3**

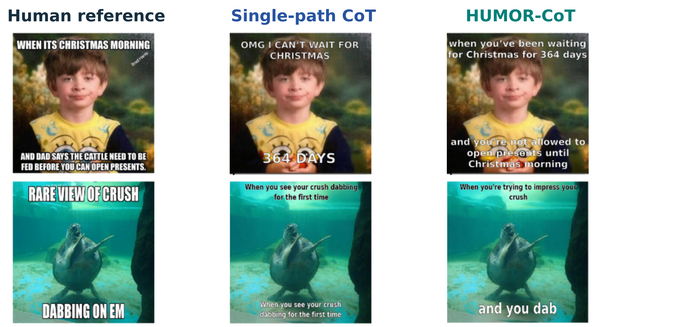
Human scores: 30 raters, 0–5. Human %: Gemini human/AI judgment.
Sim.: bge embeddings. Dist.: context-swap distance, a diversity proxy. Table 1.

Humor × context variation



CoT improves both vs. 7B. RL* is preliminary; originality falls 2.90 → 2.79.

04 How do the jokes change?



05 Does the reward agree with people?

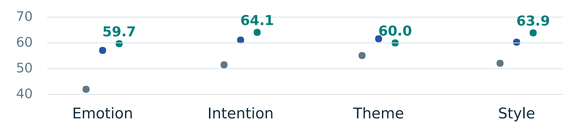
Human MaxDiff vs. reward ranks · Kendall τ ↑

Reward model / state	G1	G2	G3	G4	G5
Qwen2.5-VL-7B · base	0.16	0.28	0.47	-0.10	0.29
Qwen2.5-VL-7B · trained	0.47	0.56*	0.42	0.14	0.47
Qwen2.5-VL-32B · base	0.16	0.16	-0.02	0.14	0.29
Qwen2.5-VL-32B · trained	0.29	0.47*	0.07	0.30	0.42
Keye-VL-8B · base	0.05	0.09	0.16	0.29	0.16
Keye-VL-8B · trained	**0.78*** **	**0.77*** **	**0.78*** **	**0.78*** **	**0.78*** **

100 raters · 5 templates · *p<.05, ***p<.001 · Table 2

06 Does the meme express your intent?

Requested attributes → generated meme → VLM reclassification

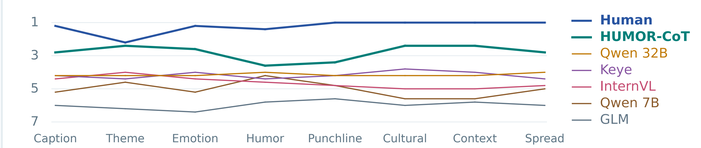


- 7B base
- 32B base
- **HUMOR 7B**

Above 32B on 3 of 4 attributes.

Table 5 · reclassification accuracy (%) · higher is better

07 Unseen templates



20 templates · Gemini-2.5-pro mean rank ↓

yingyixue7@gmail.com